

How to access the ICE cluster and run the EEG notebooks

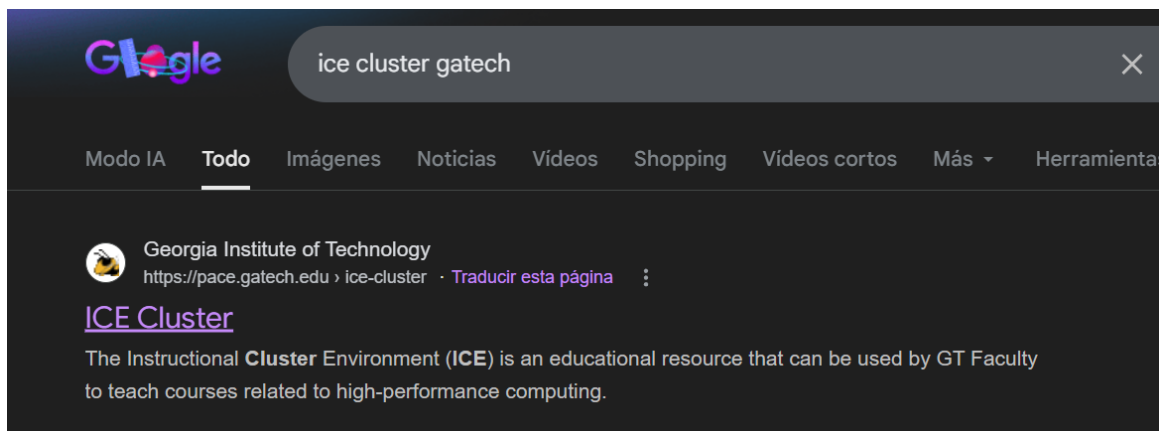
This tutorial explains how to open a VS Code session on the ICE cluster and set up the environment so you can run the EEG quickstart and advanced notebooks (*utils.py*, *student_eeg_quickstart.ipynb*, *student_eeg_advanced.ipynb*) and the subsequent analysis.

You will need your Georgia Tech login credentials, and you already have access to the ICE cluster.

Step 1: Log in to the ICE portal

Open the ICE / PACE portal in your browser and log in with your GT credentials.

<https://pace.gatech.edu/ice-cluster/>



Scroll down and press "Open OnDemand"



machines in the Linux command line is bonus practical experience for our students across all majors."

– Ethan Trewitt, M.S.
Division Chief Scientist, ICL Software Engineering and Analytics Division
Georgia Tech Research Institute (GTRI)

Request Access to ICE for Your Class

College of Computing Faculty

Faculty that is teaching a College of Computing class, or a class cross listed with this department, email your request to academicresources@cc.gatech.edu.

Other Faculty

Faculty that would like to use ICE, from a department other than the College of Computing, should to fill out this [form](#) to send an application.

[Getting Started on ICE](#) [ICE Resources](#) [Open OnDemand](#)



 Georgia Tech.

 Agent Portal | [Home](#) ▾ | [Knowledge](#) ▾ | [My Requests](#) 0 | [My Approvals](#) 0 | [System Status](#)

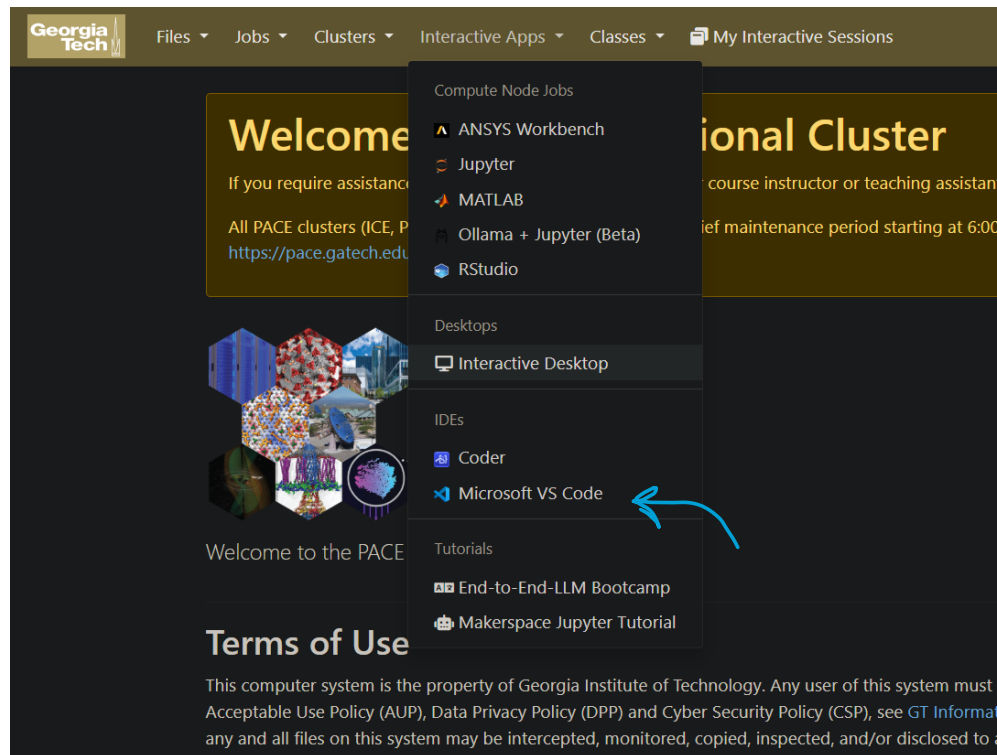
Cluster	Link
Phoenix	Phoenix OnDemand
ICE	ICE OnDemand



You will need to use the Georgia Tech VPN if you are not on campus.

Step 2: Open Interactive Apps and start VS Code

In the left menu, go to Interactive Apps. Under IDEs (or Compute Node Jobs), click Microsoft VS Code. You may see a form to configure your session.



Step 3: Configure the session

On the form, use the following exact settings to ensure you have enough power for EEG processing:

- **Quality of Service:** Select pace-ice (or the one assigned to your specific course like coc-ice).
- **Node Type:** Select CPU only, first available.
- **Number of cores (CPUs):** Enter 4.
- **Memory (GB):** Enter 16.
- **Number of hours:** Enter 1 or 2.

To ensure MNE and Jupyter work correctly, you must not use the "Default" setup.

1. In the **Environment setup** section, select **Custom environment setup....**
2. In the text box that appears, paste these exact lines:

```
module load miniforge/24.3.0-0
```

```
mamba activate base
```

```
pip install --user mne jupyter ipykernel ipynb
```



`python -m ipykernel install --user --name eeg-env --display-name "EEG_Kernel"`

3. Click **Launch** (or Submit) at the bottom of the form.

Microsoft VS Code

This app starts a browser-based Microsoft VS Code session on an ICE compute node. In the session, you are able utilize the cluster's filesystem, hardware, and software resources. See [these docs](#) for more info about this app.

VS Code version

1.99.2

Environment setup

Custom environment setup...

Default modules

No modules

Custom environment setup...

Custom Apptainer image...

- **Custom environment setup...**: Run arbitrary bash commands, which you can specify below
- **Custom Apptainer image...**: Run VS Code from an Apptainer image that you provide. The image must have VS Code installed; see [these docs](#) for info about building an image with VS Code.

Custom environment setup

```
python -m ipykernel install --user --name eeg-env --display-name "EEG_Kernel"
```

You can use enter multiple lines with arbitrary bash commands, such as module loads

Memory (GB)

16

Number of hours

4

Your job will terminate after this time limit.

- Jobs with shorter time limits are typically scheduled sooner.
- Your job will keep running even if you close your browser or lose your internet connection. You can then re-connect to the same session later, if the time limit has not expired.
- Your account will be charged while your job is running (even if it is not open in your browser). If you are done with your work before your time expires, you can cancel the job and avoid being charged for unused time.

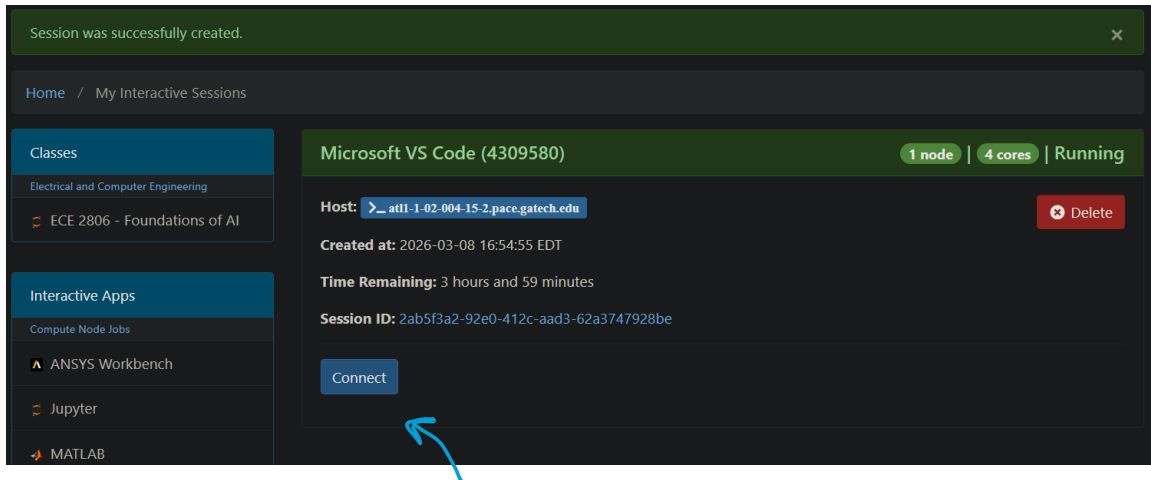
☐ I would like to receive an email when the session starts

Launch

* The Microsoft VS Code session data for this session can be accessed under the [data root directory](#).

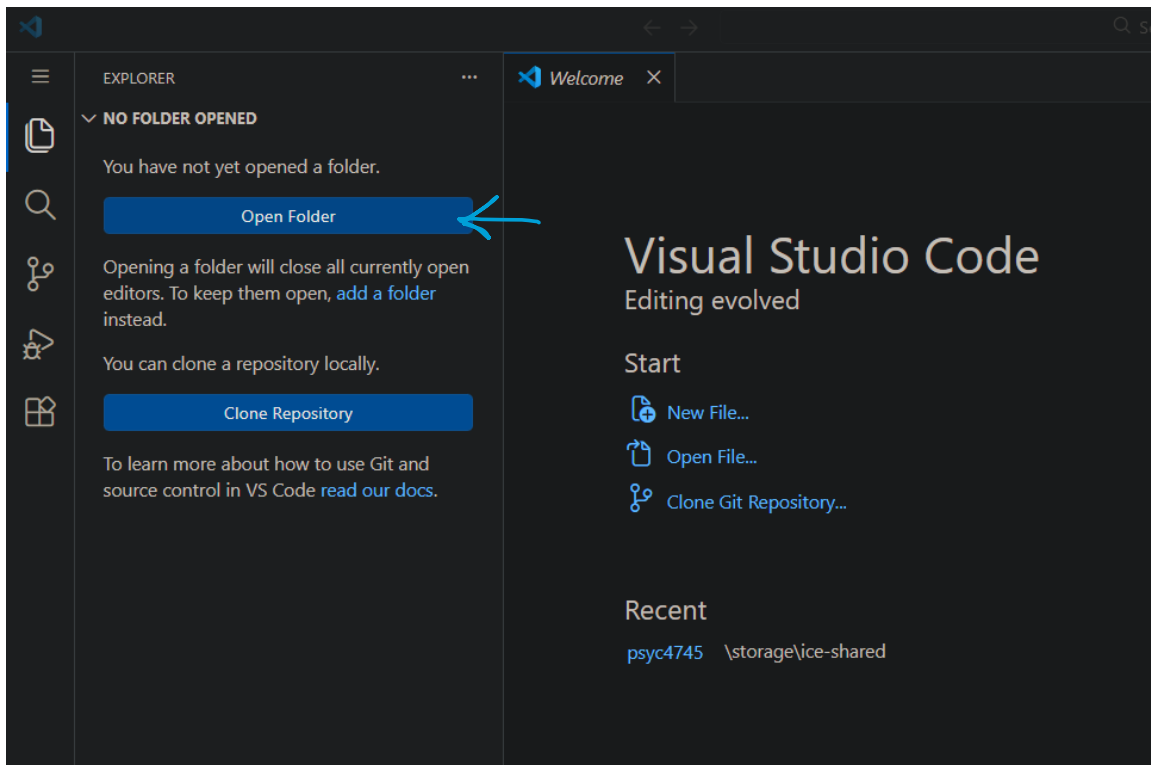
Step 5: Wait for the session and open VS Code

Your job will be queued. When it starts, a link or button to open VS Code in the browser will appear. Click it to open the VS Code session on the compute node.

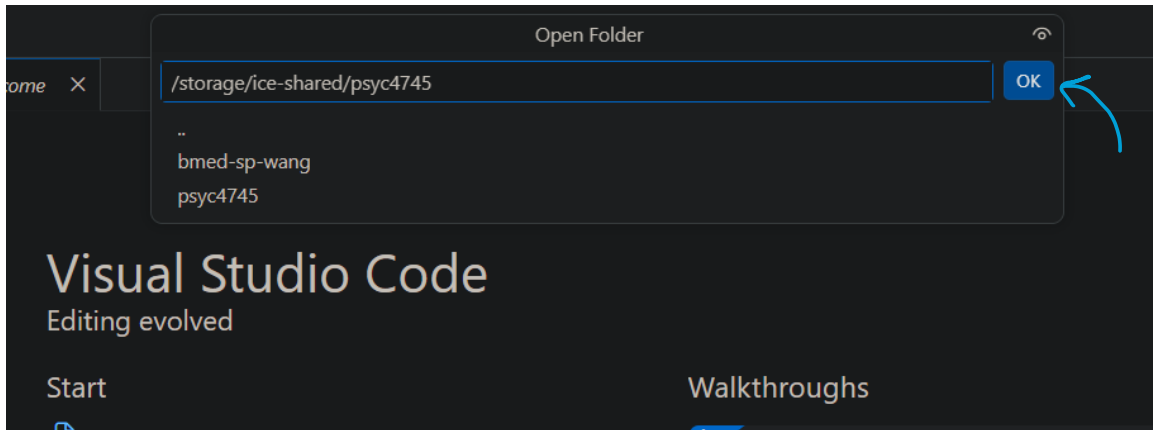


Step 6: Open the project and run the notebooks

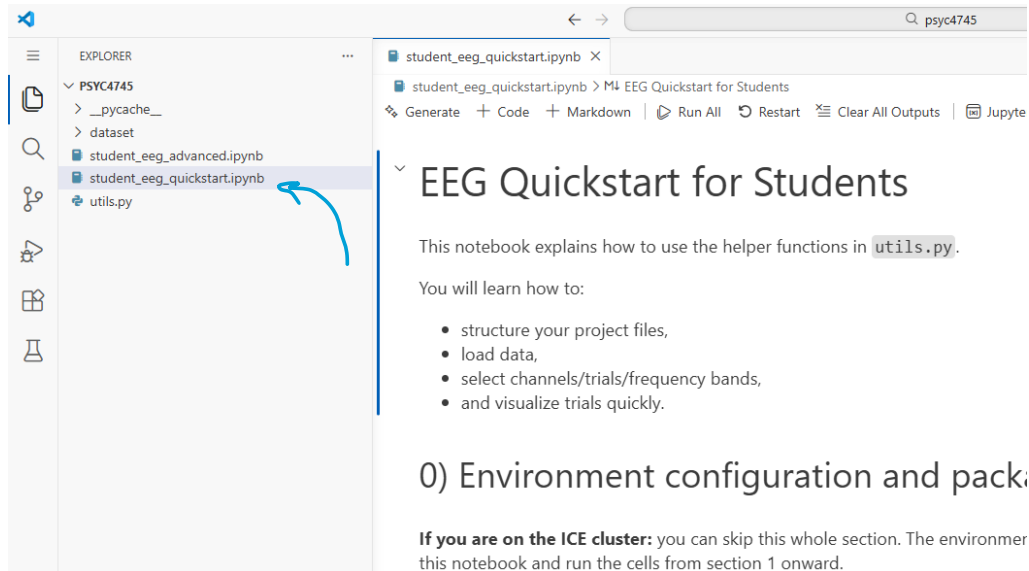
In VS Code, open the folder where the repo is. Open `student_eeg_quickstart.ipynb`, select the Python kernel (e.g. the one from the environment you activated), and run the cells. You can then open `student_eeg_advanced.ipynb` the same way.

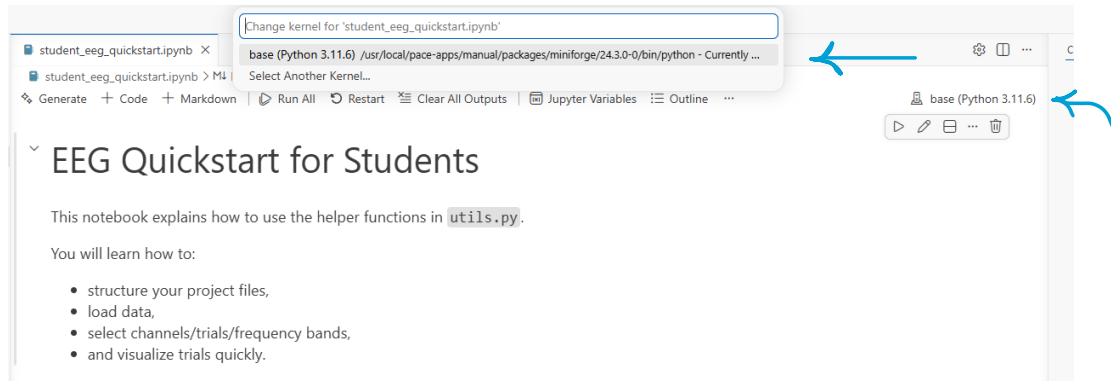


Write the path: `"/storage/ice-shared/psyc4745"` and press OK.



1. Open your .ipynb file.
2. In the top-right corner, click **Select Kernel -> Python Environments...**
3. Click the **Refresh icon** (circular arrow) and wait 10 seconds.
4. Select the one labeled **"EEG_Kernel"**.
5. If you need interactive plots, ensure the first cell of your notebook contains `%matplotlib widget`.
6. **IMPORTANT:** When VS Code opens, look for a notification at the bottom right asking to install the **"Python"** extension. Click **Install**.





Troubleshooting: "I can't see the EEG_Kernel"

If the kernel doesn't appear after refreshing:

1. Go to **Terminal -> New Terminal** inside VS Code.

2. Paste this line and hit Enter:

```
module load miniforge/24.3.0-0
mamba activate base
pip install --user mne jupyter ipykernel ipynb
python -m ipykernel install --user --name eeg-env --display-name "EEG_Kernel"
```

3. Refresh the kernel list again.